Huawei Access Node KR-ACC-02 QoS Congestion Event SOP

Node: KR-ACC-02 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_5569

Description

This Standard Operating Procedure (SOP) addresses a Major severity QoS congestion event on Huawei Access Node KR-ACC-02, specifically triggered by Microbursts, interface tail drops, and QoS queue starvation. The underlying cause of this failure domain lies in the complex interaction between the Huawei AR series router's QoS mechanism and the physical interface's bandwidth limitations. The router's QoS mechanism prioritizes traffic based on the DiffServ Code Point (DSCP) values, which may lead to queue congestion and subsequent tail drops if the interface's bandwidth is insufficient to handle the incoming traffic. Furthermore, the Huawei AR series router's QoS mechanism employs a hierarchical scheduling algorithm that allocates bandwidth to different traffic classes based on their priority and burst size, which may exacerbate queue congestion if not properly configured. The root cause of this failure domain is further complicated by the presence of Microbursts, which are high-frequency bursts of traffic that can overwhelm the interface's bandwidth and trigger congestion events despite the QoS mechanism's best efforts to mitigate them.

Pre-requisite Checks

- Check: Verify the redundant link status is stable and functioning as expected

- Check: Ensure the remote hands are dispatched to the KR Puram Access site for on-site assistance

- Check: Verify the Huawei AR series router's software version is up-to-date and patched against known QoS-related issues

Potential Root Causes

Cause	Probability
Inadequate QoS configuration	High
Interface bandwidth limitations	Medium
Microbursts and traffic bursts	Low

Diagnostic Commands & Expected Outputs

```
Command: display qos queue statistics, Expected Output: Queue statistics for all QoS queues, Description: This command will display the current queue statistics for all QoS queues on the Huawei AR series router, including the number of packets transmitted, dropped, and delayed, as well as the average queue length and delay. This will help identify if there are any queues experiencing congestion and which traffic classes are being impacted.
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```
Command: display interface statistics, Expected Output: Interface statistics for all physical interfaces, Description: This command will display the current interface statistics for all physical interfaces on the Huawei AR series router, including the number of packets transmitted, received, and dropped, as well as the interface's bandwidth utilization and errors. This will help identify if the interface's bandwidth is being saturated and if there are any physical issues affecting the interface's performance.

Command: display qos policy statistics, Expected Output: QoS policy statistics, Description: This command will display the current QoS policy statistics for all QoS policies on the Huawei AR series router, including the number of packets transmitted, dropped, and delayed, as well as the QoS policy's bandwidth allocation and scheduling algorithm. This will help identify if the QoS policy is properly configured and if there are any issues with the QoS policy's implementation.
```

Resolution Steps

Step 1: Step: Update the QoS configuration, Command: qos policy update , Description: Update the QoS policy configuration to address the identified issues, such as increasing the bandwidth allocation for impacted traffic classes or modifying the QoS policy's scheduling algorithm to better handle Microbursts.

Step 2: Step: Adjust the interface bandwidth allocation, Command: interface bandwidth , Description: Adjust the interface bandwidth allocation to ensure it can handle the incoming traffic and prevent congestion events. This may involve increasing the interface's bandwidth or modifying the QoS policy's bandwidth allocation to better match the interface's available bandwidth.

Step 3: Step: Implement traffic shaping and policing, Command: qos policy traffic-shaping , Description: Implement traffic shaping and policing to prevent Microbursts and traffic bursts from overwhelming the interface's bandwidth and triggering congestion events. This will involve configuring a traffic shaper to limit the burst size of incoming traffic and a policer to enforce the QoS policy's bandwidth allocation.

Post-Check Verification

```
Command: display qos queue statistics, Expected Output: Queue statistics for all QoS queues, Description: Verify that the queue statistics for all QoS queues have returned to normal and that the queue congestion has been resolved.

Command: display interface statistics, Expected Output: Interface statistics for all physical interfaces, Description: Verify that the interface statistics for all physical interfaces have returned to normal and that the interface's bandwidth is no longer saturated.

Command: display qos policy statistics, Expected Output: QoS policy statistics, Description: Verify that the QoS policy statistics for all QoS policies have returned to normal and that the QoS policy is properly configured and functioning as expected.
```

Roll-back Procedures

- Step: Restore the original QoS configuration, Command: qos policy restore , Description: Restore the original QoS configuration to its previous state in case the fix fails.

- Step: Adjust the interface bandwidth allocation, Command: interface bandwidth , Description: Adjust the interface bandwidth allocation to its original state in case the fix fails.

- Step: Remove traffic shaping and policing, Command: qos policy traffic-shaping remove, Description: Remove traffic shaping and policing to prevent Microbursts and traffic bursts from overwhelming the

interface's bandwidth and triggering congestion events in case the fix fails.